# FIUS-MoveSense — Functional Diagram

Ultrasonic Movement Detection System with Direction Detection

## HARDWARE LAYER

**FIUS** Sensor → **Red Pitaya** ADC Board

40 kHz ultrasonic pulse
50,000 samples/scan
~1 MHz sampling rate

CSV →

## DATA LAYER

**no_movement/** 24 files, 15K scans

**movement/** 19 files, 10.5K scans

**test/** T1-T10 files

Total: 25,490 scans
43 recording files

## TRAINING PIPELINE — main.py

**STAGE 1 — Data Loading**
data_loading.py
Read CSV + labels

→ **STAGE 2 — Signal Processing**
signal_processing.py
Bessel + Hilbert + Peak

→ **STAGE 3 — Feature Extraction**
feature_extraction.py
38 features per scan

→ **STAGE 4 — Model Training**
evaluation.py
GridSearchCV + 4 models

→ **STAGE 5 — Evaluation**
evaluation.py
Accuracy, F1, Confusion

**config.py**
Signal: 50,000
Header: 5
5 echo zones

**Time-Domain (12)**
mean, std, rms, kurtosis
energy, peak_to_peak
crest_factor, zcr, pctiles

**Zone-Based (10)**
5 zones x (mean + std)
near, close, mid, far, bg
0-50,000 sample range

**Frequency (6)**
dominant_freq, mean_freq
spectral_entropy, flatness
band_energy, centroid

**Temporal (9) □**
scan-to-scan differences
correlation, energy_diff
zone diffs (KEY INNOVATION)

**4 Classifiers**
Random Forest □ 90.5%
SVM 88.7%
KNN 85.1% | LR 79.5%

**Saved Artifacts**
*.joblib (4 models)
scaler.joblib
feature_order.txt

## PREDICTION PIPELINE — predict.py

**New CSV File**
Any encoding

→ **Filter + Envelope**
Bessel + Hilbert
Peak detection

→ **Extract 38**
Features
+ Scale

→ **Classify**
predict_proba ≥ 0.65
→ MOVING / NOT MOVING

→ **Direction Detection**
Envelope peak trend
→ / ← / ↔

→ **OUTPUT**
□ Movement DETECTED
→ APPROACHING sensor
71.5% confidence | ▇▇▇